AI-Based Detection and Tracking of Cardiac Interventional Devices

1. Problem Statement

During cardiac interventional procedures, clinicians use live medical images to guide, position and deploy multiple devices inside the cardiovascular system. These devices may include:

- Coronary stents
- Balloon catheters
- Guidewires and guide catheters
- Atherectomy devices
- Pacemaker leads and implanted pacemakers
- Artificial or prosthetic cardiac valves
- Transcatheter aortic valve devices
- Transcatheter mitral valve devices

Detecting these devices manually from fluoroscopy and angiography images is challenging because the devices are small, have low visual contrast, may overlap with anatomical structures and other instruments, and can move rapidly due to cardiac motion, respiration and clinician manipulation.

The objective is to develop an AI-based medical imaging solution that can automatically detect, classify, localize and track cardiac interventional devices in real time or near real time during cardiac procedures.

The solution should help clinicians understand:

1. Which devices are currently visible in the image.
2. Where each device or device component is located.
3. How the device is moving across consecutive frames.
4. Whether the device is positioned relative to the intended cardiac anatomy.
5. Whether a device has appeared, disappeared, changed state or been deployed during the procedure.

2. Primary Objective

Build a multi-device computer-vision model capable of processing cardiac X-ray fluoroscopy and angiography image sequences and producing real-time device detection, segmentation and tracking information.

The proposed system must support simultaneous detection of multiple devices and must remain reliable under low-dose, noisy and partially occluded imaging conditions.

3. Recommended Initial Scope

Primary imaging modality

The first release should focus on:

- X-ray fluoroscopy
- Coronary angiography
- Interventional X-ray image sequences
- DICOM images and DICOM video sequences

Echocardiography, intravascular ultrasound, OCT and cardiac CT may be considered in later releases.

Target device classes

Category	Devices or components to detect
Coronary intervention	Stent, stent markers and deployed stent
Balloon devices	Balloon catheter, balloon markers and inflated balloon
Catheter devices	Guide catheter, diagnostic catheter and catheter tip
Navigation devices	Guidewire and guidewire tip
Plaque-treatment devices	Rotational, orbital or laser atherectomy catheter
Electrophysiology devices	Ablation catheter, mapping catheter and intracardiac echo catheter
Cardiac rhythm devices	Pacemaker, pacemaker lead and lead tip
Structural-heart devices	Artificial valve and valve-delivery system
Aortic procedures	TAVR/TAVI valve, delivery catheter and deployed aortic valve
Mitral procedures	Transcatheter mitral valve or repair device and delivery system

4. AI Capabilities Required

The AI solution should provide the following capabilities:

Device presence detection

Identify whether each supported device type is present within an image or image sequence.

Device classification

Classify detected devices into their appropriate clinical categories, such as stent, balloon catheter, guidewire, pacemaker lead or valve-delivery device.

Device localization

Generate bounding boxes, key points or landmarks for the detected devices.

Examples include:

- Catheter tip
- Guidewire tip
- Proximal and distal balloon markers
- Stent boundaries
- Pacemaker lead tip
- Valve frame boundaries
- Valve-delivery-system markers

Device segmentation

Generate pixel-level masks for devices where their complete shape or boundaries must be visualized.

Temporal tracking

Assign a persistent tracking identifier to each device and follow it across consecutive fluoroscopy frames.

The model should continue tracking devices through:

- Temporary occlusion
- Overlap with other instruments
- Patient or table movement
- Contrast-agent injection
- Cardiac motion
- Rapid device movement

Device-state recognition

Where sufficient training data are available, classify the procedural state of the device.

Examples include:

- Balloon deflated versus inflated
- Stent undeployed versus deployed
- Valve crimped versus partially deployed versus fully deployed
- Catheter advancing, retracting or stationary
- Device entering or leaving the field of view

Confidence and uncertainty estimation

Provide a confidence score for every detection and flag uncertain predictions instead of presenting them as definitive results.

5. Expected Model Output

For each frame, the system should return:

- Device class
- Device instance ID

- Bounding box or segmentation mask
- Device landmarks or tip coordinates
- Detection confidence
- Tracking confidence
- Device state, where applicable
- Frame number and timestamp
- Warning when the device is partially visible or uncertain

Example output:


```json
{
  "frame_id": 1642,
  "devices": [
    {
      "device_class": "balloon_catheter",
      "instance_id": "device_01",
      "bounding_box": [312, 205, 482, 248],
      "landmarks": {
        "proximal_marker": [328, 226],
        "distal_marker": [465, 224]
      },
      "device_state": "inflated",
      "detection_confidence": 0.96,
      "tracking_confidence": 0.93
    }
  ]
}
```


6. Recommended Performance Targets

The following should be treated as initial engineering targets and validated with Philips clinical and product stakeholders.

Metric	Initial target
Device-presence sensitivity	≥95% for priority devices
Device-presence specificity	≥95%
Mean average precision	≥90% for priority device classes
Catheter or guidewire tip error	≤2–3 mm or clinically agreed pixel equivalent
Balloon-marker localization error	≤2 mm or clinically agreed pixel equivalent
Tracking success rate	≥95% across valid sequences
Device classification accuracy	≥95%
Processing speed	Minimum 15 FPS
Preferred real-time performance	25–30 FPS

Metric	Initial target
Per-frame inference latency	≤100 milliseconds
Uncertain-case identification	≥90% recall for low-confidence cases

Performance must also be reported separately for each device type, imaging system, procedure type, patient group and image-quality category.

7. Key Technical Challenges

The AI team must specifically address:

- Very small and thin devices occupying few pixels
- Low-dose X-ray noise
- Low contrast between the device and surrounding anatomy
- Motion caused by heartbeat and respiration
- Device overlap and occlusion
- Multiple similar-looking catheters in the same image
- Changes in imaging angle and magnification
- Presence or absence of contrast medium
- Different manufacturers and device designs
- Different radiopaque marker patterns
- Differences among Philips imaging-system configurations
- Limited examples of rare devices and procedural complications
- Domain shift across hospitals, geographies and clinical workflows

8. Data and Annotation Requirements

The training dataset should contain anonymized DICOM studies and video sequences from different:

- Hospitals and geographies
- Philips imaging-system models
- Procedure types
- Imaging angles
- Radiation-dose settings
- Adult patient anatomies
- Device manufacturers and models
- Levels of image quality
- Normal and complex procedures

Annotations should include:

- Device class
- Bounding box
- Segmentation mask, where required
- Device tip and marker landmarks
- Device instance identifier
- Device state
- Visibility and occlusion level
- Procedure phase

- Image-quality rating
- Start and end frames for each device occurrence

Annotations should be reviewed by trained clinical annotators and sampled for validation by interventional cardiologists or qualified clinical experts.

9. Proposed Solution Approach

A recommended model architecture would combine:

1. A multi-class object-detection model for identifying device presence and location.
2. A key-point detection model for device tips and radiopaque markers.
3. A segmentation model for stents, balloons, catheters and valve frames.
4. A temporal tracking model that uses information from consecutive frames.
5. A device-state classification model.
6. An uncertainty-estimation layer for clinical safety.
7. A rules or workflow engine for converting model predictions into clinically meaningful events.

The model should use both spatial information from individual frames and temporal information from the complete fluoroscopy sequence.

10. Intended Clinical and Product Value

The proposed AI solution is intended to:

- Improve device visibility during complex cardiac procedures.
- Reduce the cognitive workload on interventional clinicians.
- Support more accurate device positioning and deployment.
- Automatically identify important procedural stages.
- Enable device-focused image enhancement.
- Support automated measurements and procedural documentation.
- Reduce repeated imaging where clinically appropriate.
- Provide structured device information for downstream Philips applications.
- Create a reusable AI foundation for coronary, electrophysiology and structural-heart procedures.

11. Safety and Regulatory Boundary

The initial solution should function as a clinical decision-support and image-guidance capability. It should not independently diagnose a disease, recommend a device, control device movement or automatically deploy a device.

All predictions must:

- Be presented with confidence information.
- Allow clinician confirmation or rejection.
- Preserve the original image.
- Maintain a traceable audit log.
- Be validated across sites and device manufacturers.
- Undergo clinical, cybersecurity, privacy, usability and regulatory assessment before clinical deployment.

12. Definition of Success

The initiative will be considered successful when the model can reliably detect and track the agreed priority devices on unseen Philips fluoroscopy studies, operate within the required clinical latency, generalize across hospitals and imaging conditions, and demonstrate measurable improvement in device visualization, workflow efficiency or procedural consistency without disrupting the clinician's existing workflow.